



A Case Study Using Web Objects and COSMIC for Effort Estimation of Web Applications

Filomena Ferrucci¹, **Carmin Gravino**¹, Sergio Di Martino²

¹ Università degli Studi di Salerno

² University of Napoli “Federico II”

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Outline

- Motivation and research goals
 - Estimating web applications development effort
 - Size measures: COSMIC and Web Objects
- Research method
 - Dataset
 - Estimation technique & analysis methods
 - Empirical results
- Conclusions and future work

How estimate web application development effort?

- In the context of traditional software engineering several size measures have been proposed
 - FP, OOFP, LOC, and others
- Such traditional size measure turn out to be rather inadequate when dealing with web applications
- The commonly software engineering measures and methods have to be revised and/or adapted

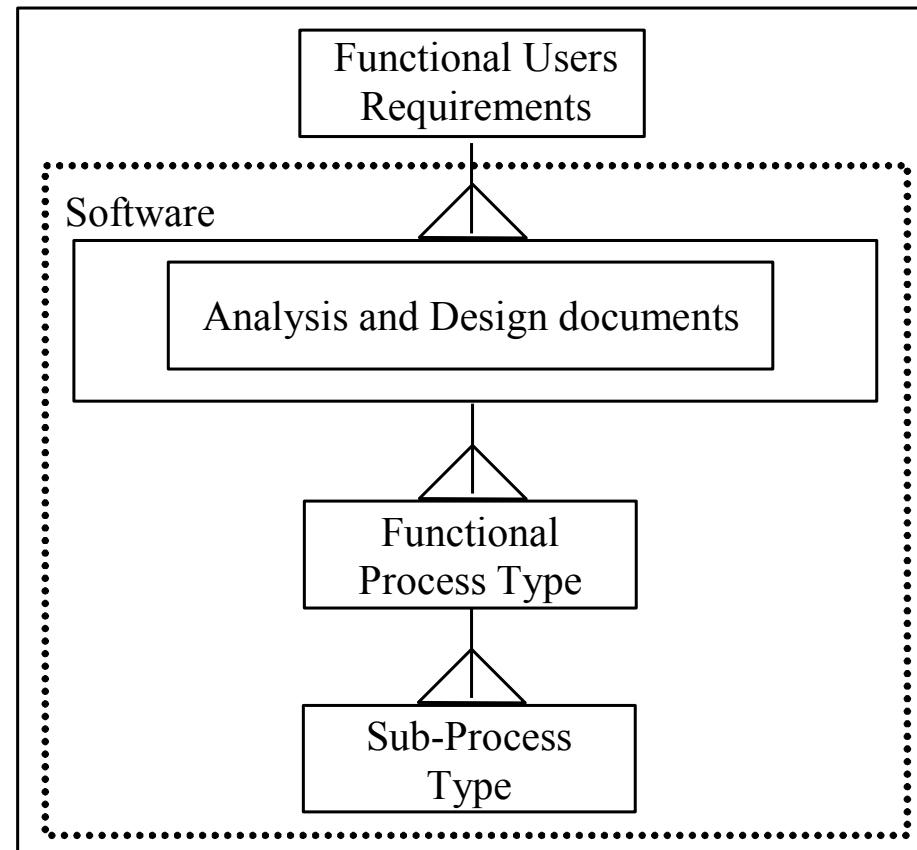
Some proposals

- ***Web Objects*** method (by Reifer) adds specific characteristics of web applications to the five function types of FPA
 - ❑ An empirical study (by Ruhe et al.) showed that Web Objects performed better than Function Points
- Rollo described the difficulties of FPA when applied to web applications
 - ❑ ***COSMIC*** resulted to be the most flexible approach for counting the functional size of web applications (no empirical study)
 - ❑ Only two studies employed COSMIC, obtaining contrasting results (Mendes et al.'s study and our previous study)
- To date, there is a strong need for further empirical evidence to support these size measures
 - ❑ ... it motivated the current study

COSMIC

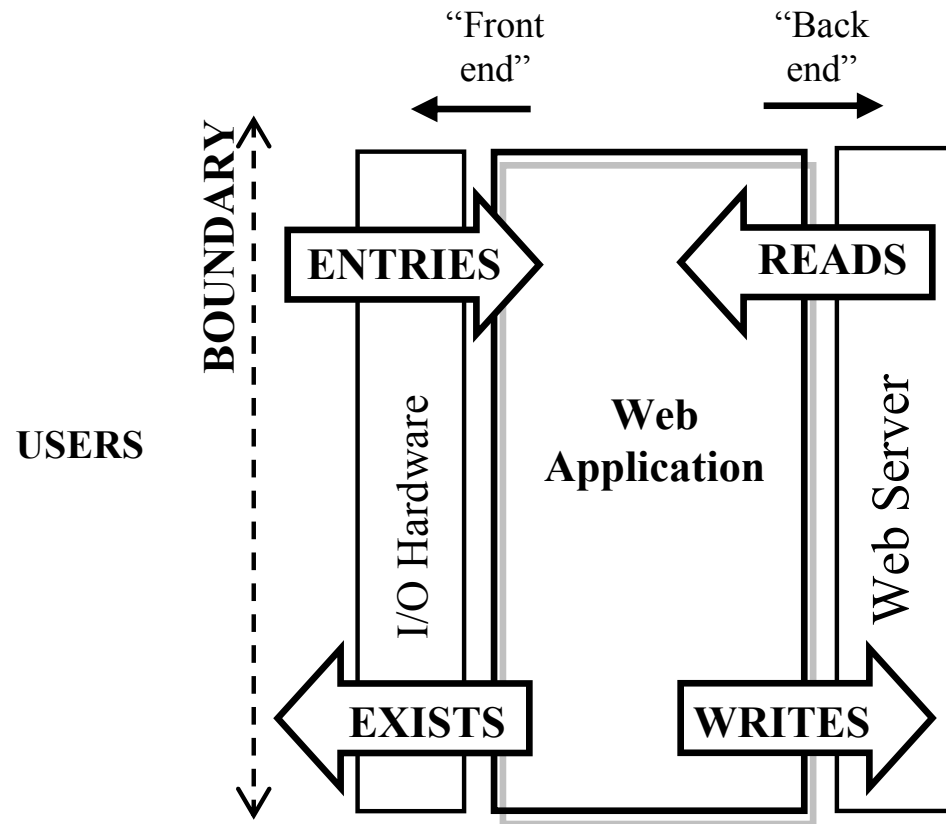
- *COSMIC* focuses on counting data movements ...
 - It assumes that software manipulates pieces of information
 - The software functional requirements are implemented by a set of functional processes
 - Each process is an ordered set of sub-processes performing either data movements or data manipulation
 - *The functional size of a software is directly proportional to the number of its data movement sub-processes*
- Web applications are characterized by a significant amount of interactions with users and data sources (e.g. databases)
 - Web applications can be suitably sized by using the principles underlying COSMIC for counting data movements

The Software Model we considered



- the data movement sub-processes are identified by analyzing analysis and design documents (e.g, use cases and class diagrams)

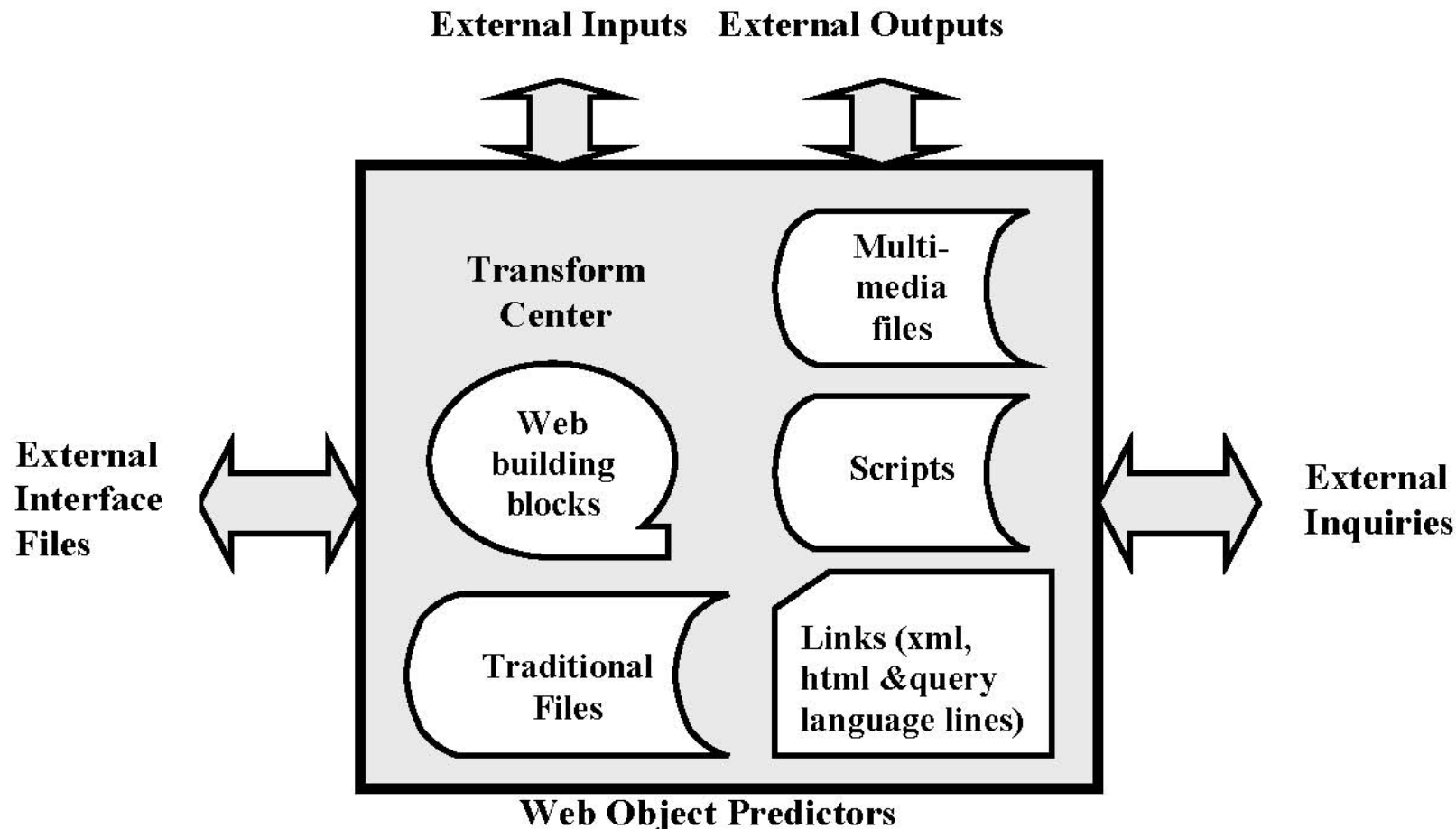
The Context Model we considered



- Unit measure: 1 CFP (*Cosmic Functional Point*), equivalent to a single data movement at each sub-process
- $\text{Size}_{\text{CFP}} = \sum(\text{ENTRY}) + \sum(\text{EXIT}) + \sum(\text{READ}) + \sum(\text{WRITE})$

Web Objects

- New four Web-related components added to the five function types of the *FPA* method



Web Objects (2)

<i>Internal Logical Files</i>	logical, persistent entities maintained by the Web application to store information of interest
<i>External Interface Files</i>	logical, persistent entities that are referenced by the Web application, but are maintained by another software application
<i>External Inputs</i>	logical, elementary business processes that cross into the application boundary to maintain the data on an Internal Logical File, access a Multi-Media File, invoke a Script, access a Link or ensure compliance with user requirements
<i>External Outputs</i>	logical, elementary business processes that result in data leaving the application boundary to meet a user requirements (e.g., reports, screens)
<i>External Queries</i>	logical, elementary business processes that consist of a data “trigger” followed by a retrieval of data that leaves the application boundary (e.g., browsing of data)
<i>Multi-Media Files</i>	physical, persistent entities used by the Web application to generate output in multi-media format.
<i>Web Building Blocks</i>	logical persistent entities used to build the Web applications and automate their functionality
<i>Scripts</i>	logical, persistent entities used by the Web application to link internal files and building blocks together in predefined patterns
<i>Links</i>	logical, persistent entities maintained by the Web application to find links of interest to external applications

Web Objects (3)

- To apply the *Web Objects* method
 - ❑ First, the instances of the nine components are counted and a complexity (low, average, or high) is associated to them
 - ❑ Then, a weight is associated to each counted instance of the components, by using a calculation worksheet provided by Reifer
 - ❑ The total amount of *Web Objects* is obtained according to the weights that are related to the complexity of each component

Web Objects (4)

Web Object Predictors	Low	Average	High	Notes
# function, object or application points	*	*	*	Doesn't matter so long as one predictor is selected and is consistently counted in unadjusted form (include query & hypertext languages)
- Internal Logical Files - External Interface Files - External Inputs - External Outputs - External Inquires	7 5 3 4 3	10 7 4 5 4	15 10 6 7 6	Unadjusted function points computed using these standard weightings using IFUG counting conventions
# multi-media files	4	5	7	Text, video, audio, etc. (including graphics files, pictures, images, etc.)
# web building blocks	3	4	6	Shopping carts, widgets, components (includes COTS components), active components (applets, agents, etc.)
# scripts (animation, audio, video, visual, etc.)	2	3	4	Macros, containers, etc.
# of links (xml, html and query language lines)	3	4	6	Logical line counts, not physical

* Weighted per the standards for the counting convention adopted

Web Object Calculation Worksheet

Related work

- Ruhe et al. study:
 - Web Objects, OLS technique and a dataset of 12 web applications of an Australian software company
- Mendes et al.'s study
 - COSMIC, OLS, and a dataset of 37 hypermedia systems (no web apps), developed by academic students counting based on the final implementation
- Our previous study:
 - COSMIC, OLS and a dataset of 44 web applications, developed by academic students

The goals of our case study

- Research questions:

RSQ1 How good is a COSMIC based model to estimate web application development effort?

RSQ2 How good is a Web Objects based model to estimate web application development effort?

RSQ3 Which, between COSMIC and Web Objects, provide better results?

Outline

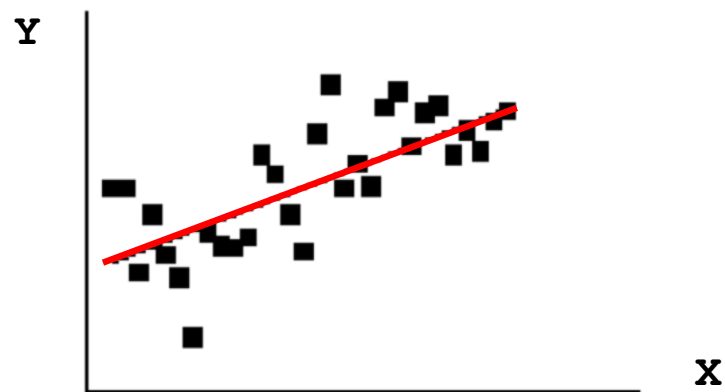
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Dataset

- Availability of reliable project data coming from the industrial world is a crucial aspect ...
- Data for our study provided by an Italian software company
 - specialized in the design, development, and management of solutions for Web portals, Geographical Information Systems, etc.
 - certified ISO 9001:2000, and also a certified partner of Microsoft, Oracle, and ESRI
- The company provided us two sets of 15 and 4 Web applications
 - The applications of the second set developed after the first 15
 - e.g., e-government, e-banking, Web portals, and Intranet applications
 - Developed by exploiting a wide range of Web-oriented technologies, such as J2EE, ASP.NET, etc.

Estimation technique: OLS Regression

- Regression modeling is used to fit a dependent variable, Y , as a function of one or more independent variables (predictors), X_1 - X_n
 - This technique fits the data to the specified model trying to minimize the overall sum of the squared errors



- We used development effort (EFH) as a dependent variable and size measured in terms of COSMIC (CFP) (or Web Objects (WO)) as an independent variable
 - $EFH = a + b * CFP$ (the CFP model)

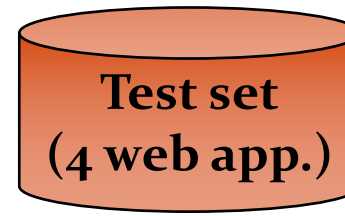
Hold-out cross validation

- **Validation**, using two different datasets:
 - A training set to build the estimation model
 - A test set to validate the estimation model

RQ1



COSMIC
based
model



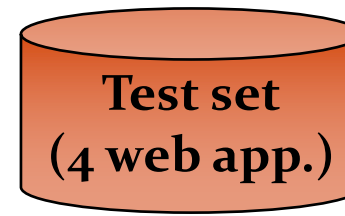
Prediction
Accuracy

RQ3

RQ2



Web Objects
based
model



Prediction
Accuracy

- This is (theoretically) the best option since the test/validation set is a totally separate dataset from the dataset used to build the effort estimation models

Assessing the accuracy of the estimations

- *Mean & Median of Magnitude of Relative Error (MMRE & MdMRE)*

$$MRE = |EFH_{real} - EFH_{pred}| / EFH_{real}$$

to measure the aggregation of *MRE* over the observations in the test set

- An *MMRE* less than 0.25 (i.e., 25%) can be considered acceptable (Conte et al. threshold)

- The *Prediction at level l*, $Pred(0.25) = k / N$

- A *Pred(0.25)* greater than 0.75 (i.e., 75%), meaning that more than 75% of the estimated values fall within 25% of their actual values, can be considered acceptable (Conte et al. threshold)

- *Boxplots of absolute residuals ($|EFH_{real} - EFH_{pred}|$)*

Results

The obtained models

		Value	Std.Err	t-value	p-value
CFP	Coefficient	3.014	0.386	7.801	0.000
	Intercept	862.281	250.613	3.441	0.004
WO	Coefficient	1.246	0.241	5.162	0.000
	Intercept	851.912	375.814	2.267	0.041
	R^2	Adjusted R^2	Std. Err	F	Sign. F
CFP	0.824	0.810	360.099	60.861	0.000
WO	0.672	0.647	491.513	26.645	0.000

■ CFP model:

- $R^2=0.824$ indicates that 82.4% is the amount of the variance of the dependent variable *EFH* that is explained by the model
- The high F value = 60.861 and the low p -value=0.000 indicate that a prediction is indeed possible with a high degree of confidence

■ WO model:

- The R^2 can be considered fair and the F value is not so high, however the regression equation can be considered significant (p -value=0.000)

Results (2)

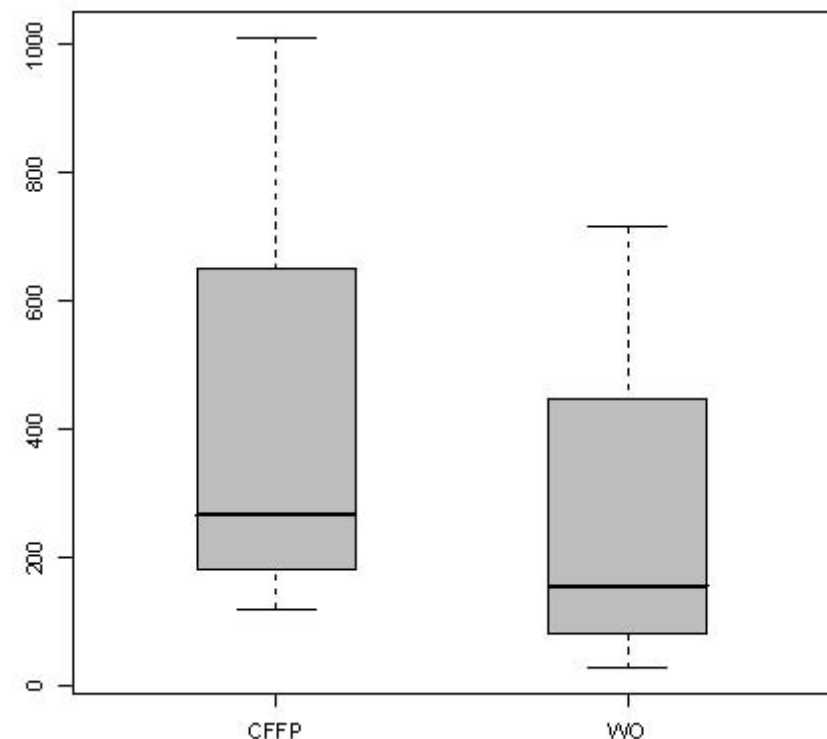
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- For both the models the *p-value* of coefficients and intercepts are less than 0.05, so they are significant predictors with a confidence of 5%
- As for the *t-value*, a variable is significant if the corresponding *t-value* is greater than 1.5, which is the case for both the models

Results (3)

- Although the CFP model fits better than the WO model
 - (i.e., better R^2 and F values)
- MMRE and boxplots of absolute residuals show better performance for the WO model
- However, no statistically significant difference for the absolute residuals (Wilcoxon test)
- ... the measures have comparable prediction accuracies

	MMRE	MdMRE	Pred(0.25)
CFP	0.22	0.11	0.75
WO	0.14	0.06	0.75



Comparison with related work

- Ruhe et al. study:
 - Web Objects, OLS technique and a dataset of 12 web applications of an Australian software company → $MMRE = 0.24$ and $Pred(0.25) = 0.75$
 - our results confirm the goodness of WO and extend the Ruhe et al. results:
 - $mean(EFH \text{ of our study}) \text{ about } 3 * mean(EFH \text{ of Ruhe et al.'s study})$
 - $mean(WO \text{ of our study}) \text{ about } 5 * mean(WO \text{ of Ruhe et al.'s study})$
- Mendes et al.'s study
 - COSMIC, OLS, and a dataset of 37 hypermedia systems (no web apps), developed by academic students → No positive results
 - counting based on the final implementation
- Our previous study:
 - COSMIC, OLS and a dataset of 44 web applications, developed by academic students → $MMRE = 0.15$ and $Pred(0.25) = 0.75$
 - current study with the industrial dataset confirms that COSMIC can be effectively employed to estimate Web application development effort

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Conclusions

- *COSMIC* and *Web Objects*
 - adopted to measure functional size of dynamic web applications
 - ... and applied on analysis and design documents
 - employed with OLS to provide early effort estimations
- We obtained interesting results for the software company that supplied the data
 - they might be also relevant for other companies that develop Web applications similar to those used in our investigation
 - However, a replication of the study, in different setting and with a larger dataset, is required in order to generalize the results to other companies